Lab - Shared Access Signatures (SAS) - At the Container Level

To Begin with the Lab

Summary

This lab demonstrates how to generate a **Container-level Shared Access Signature (SAS)** and use it in **Azure Storage Explorer**. A Container SAS provides secure, temporary, and permission-based access to all blobs within a specific container without exposing the storage account access keys.

Understanding

A **Container SAS** grants access to an entire blob container rather than a single blob. It allows you to specify permissions such as **Read, Write, Delete, List, Create, and Add**, along with a **start time** and **expiry time**. The SAS is signed using the storage account key (or a user delegation key) and can be used to securely connect tools like **Azure Storage Explorer**. Unlike a **Blob SAS**, which provides access to only one file, a **Container SAS** allows users to work with multiple blobs inside the container according to the permissions assigned.

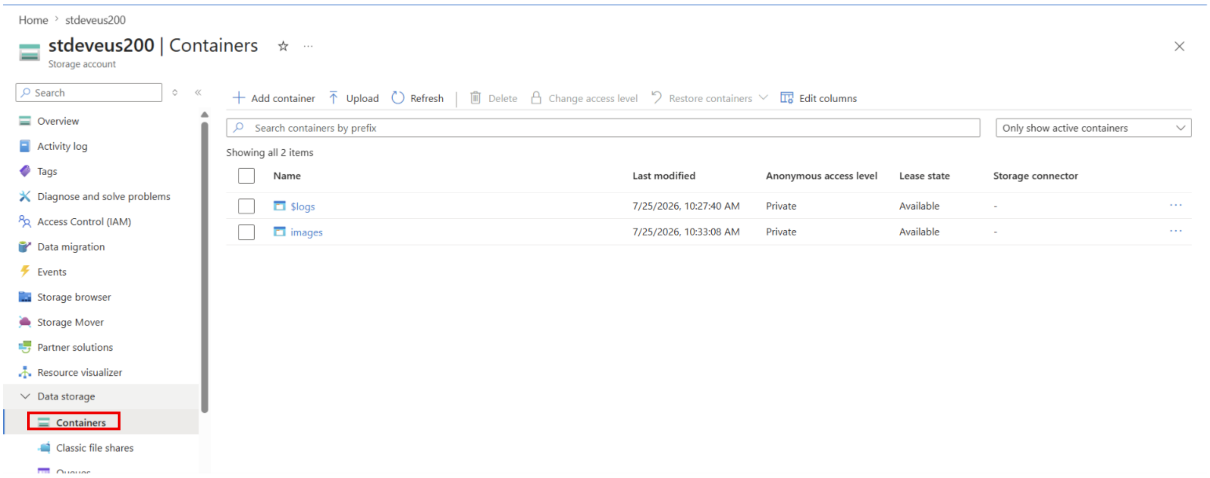
Example:

Suppose you want a team member to upload images to the **images** container for one week. Instead of sharing the storage account access key, you generate a **Container SAS** with **Read, Write, List, and Create** permissions that expires after seven days. The team member connects to the container using Azure Storage Explorer and can only perform the allowed operations during that time.

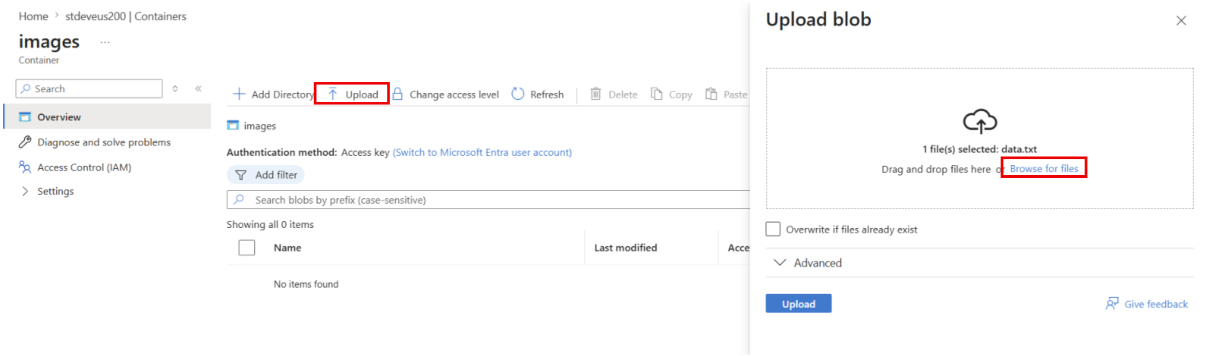
Lab Steps

Step 1: Open the Storage Account

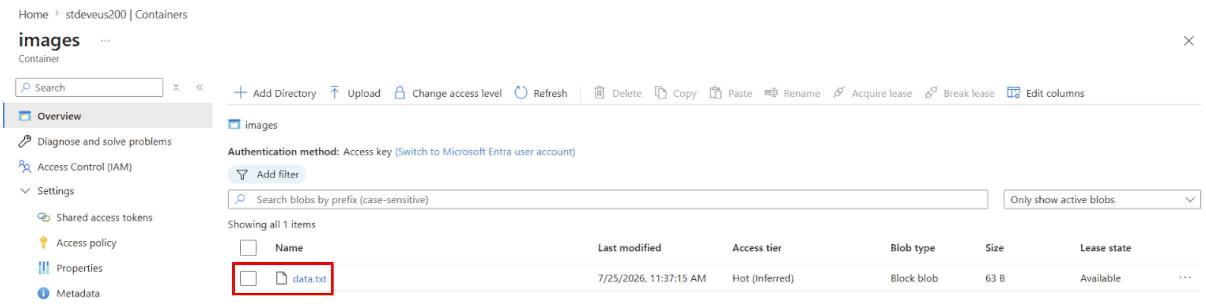
- Sign in to the Azure Portal.
- Open your **Storage Account**.
- Select **Containers**.



- Open the container(**images**).
- Click on the **Upload**.
- Click on the **Browse for File.**
- Choose **any File.**

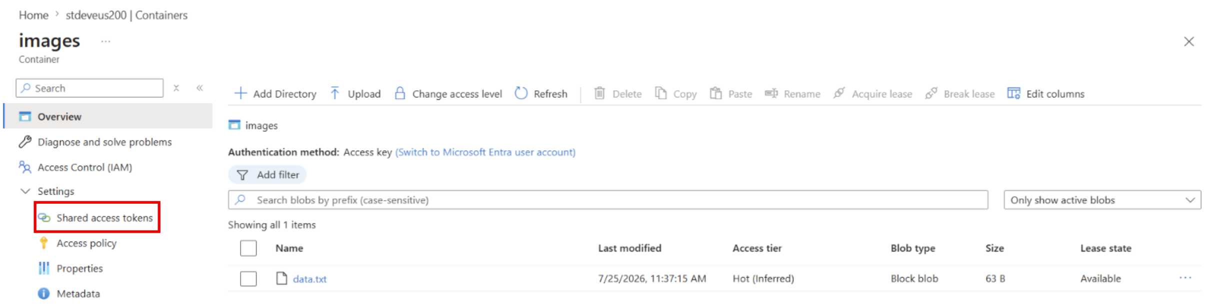


- Uploaded File.

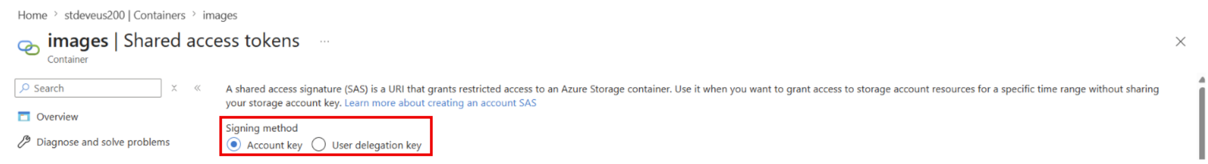


Step 2: Generate a Container SAS

- Select the container(**images**).
- Click on the **Shared access tokens (left menu).**

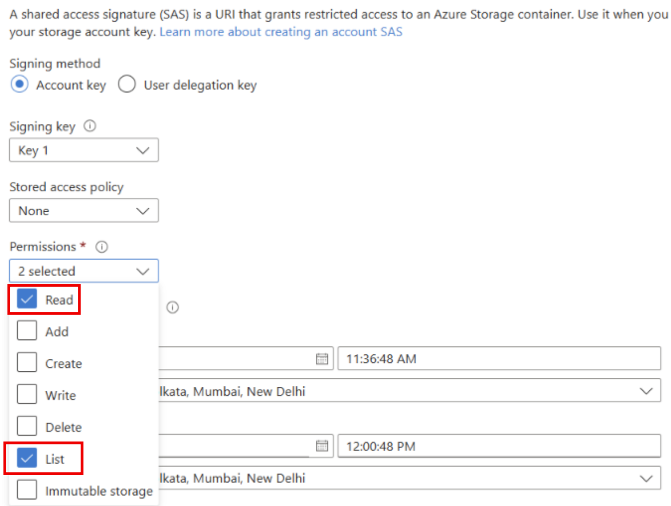


- Choose the **Signing Method** as **Account Key**.



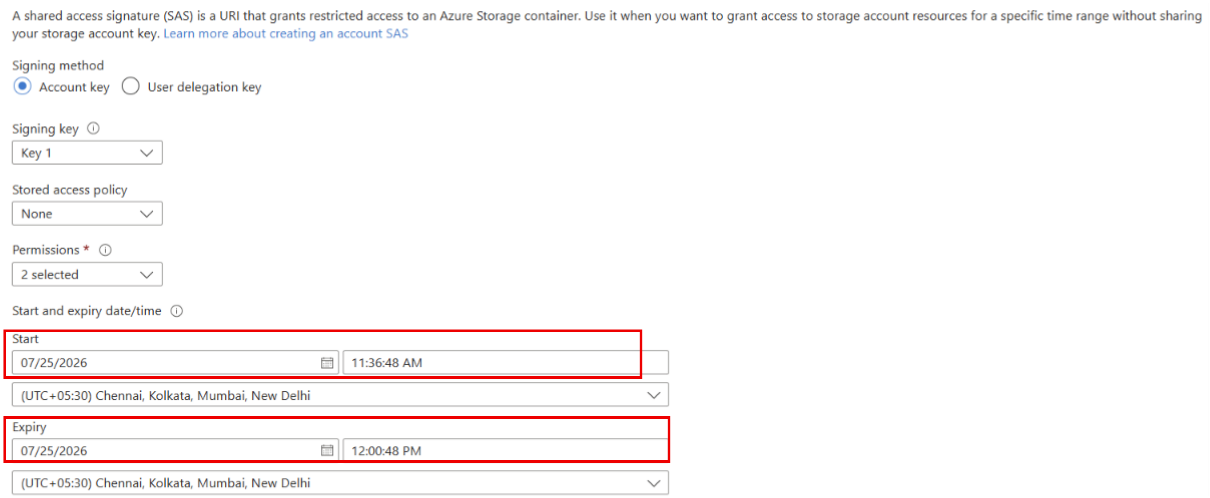
Step 3: Configure Permissions

- Select the permissions required for the container.
- Common permissions include:
  - Read
  - List
- Grant only the permissions that are necessary.



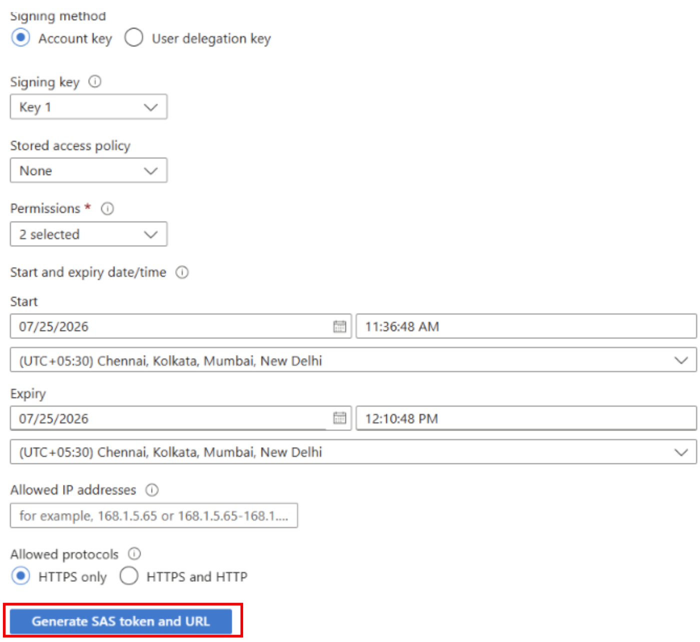
Step 4: Set the Validity Period

- Specify the **Start Date and Time**.
- Specify the **Expiry Date and Time**.
- The SAS will only be valid during this period.



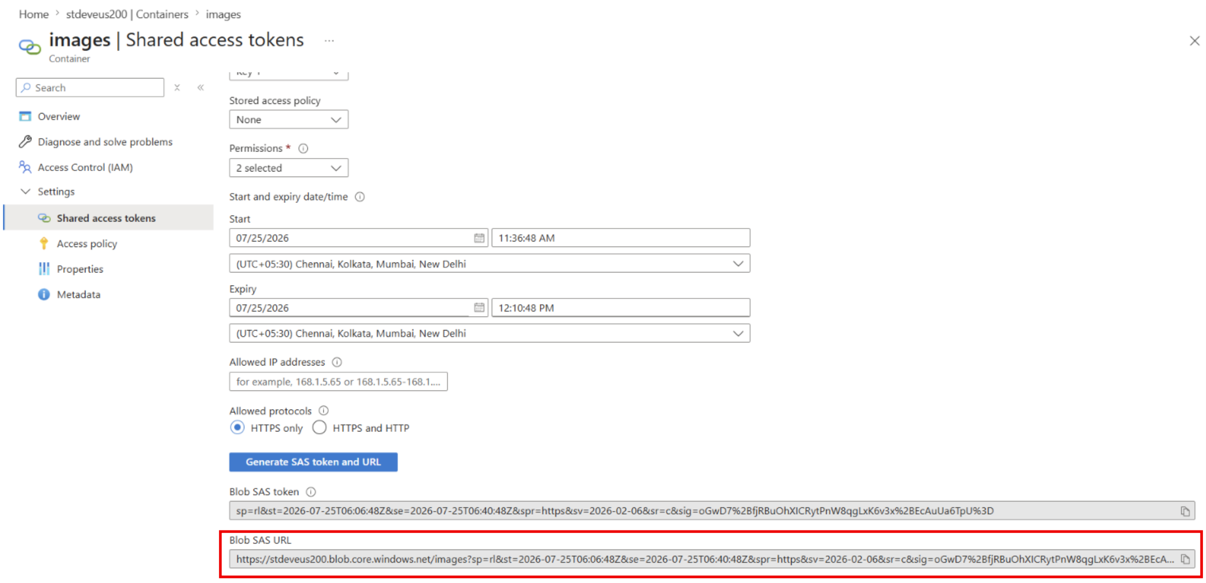
Step 5: Generate the SAS URL

- Click **Generate SAS Token and URL**.
- Azure generates:
  - Container SAS Token
  - Container SAS URL



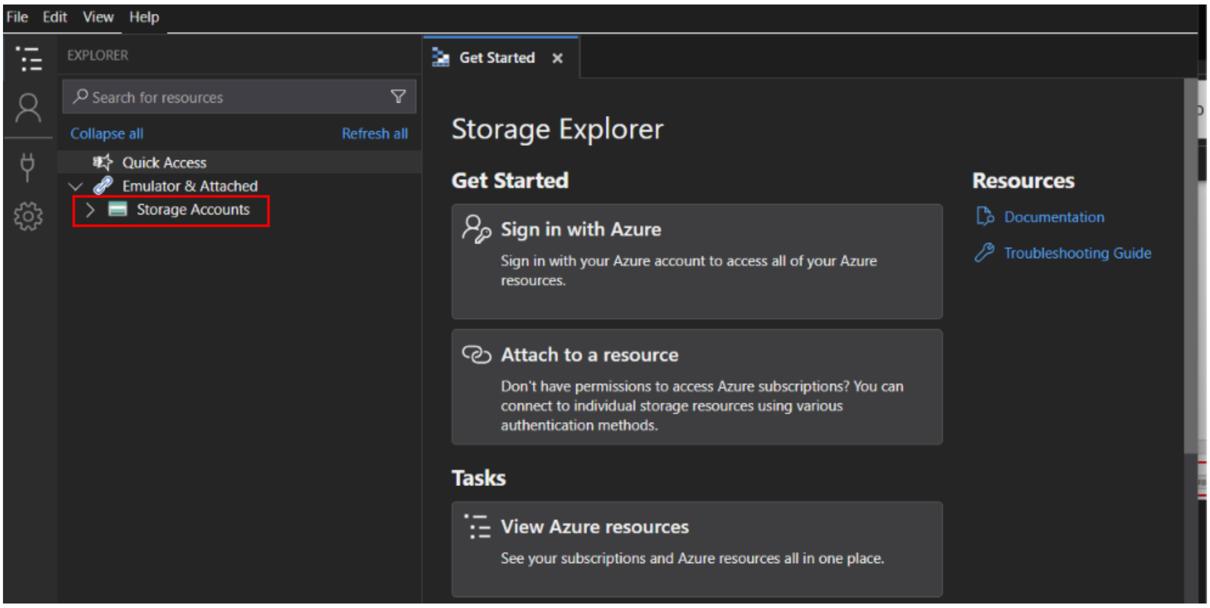
Step 6: Copy the Container SAS URL

- Copy the generated **Container SAS URL**.
- Keep it ready for use in Azure Storage Explorer.

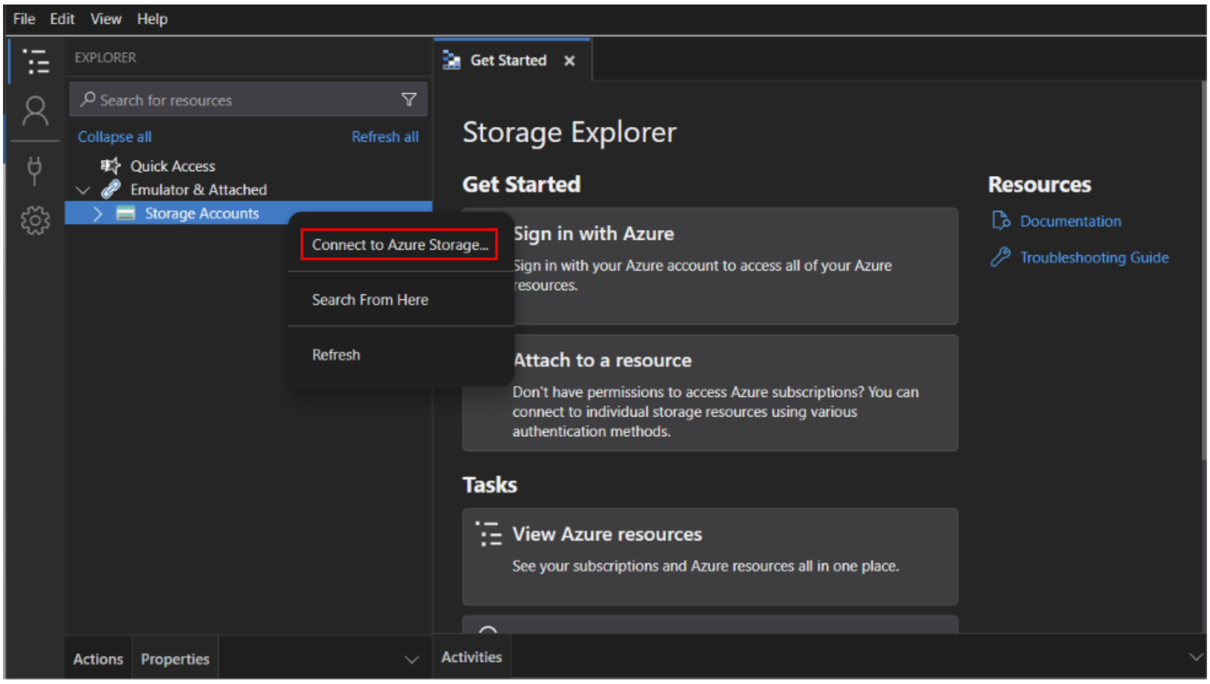


Step 7: Open Azure Storage Explorer

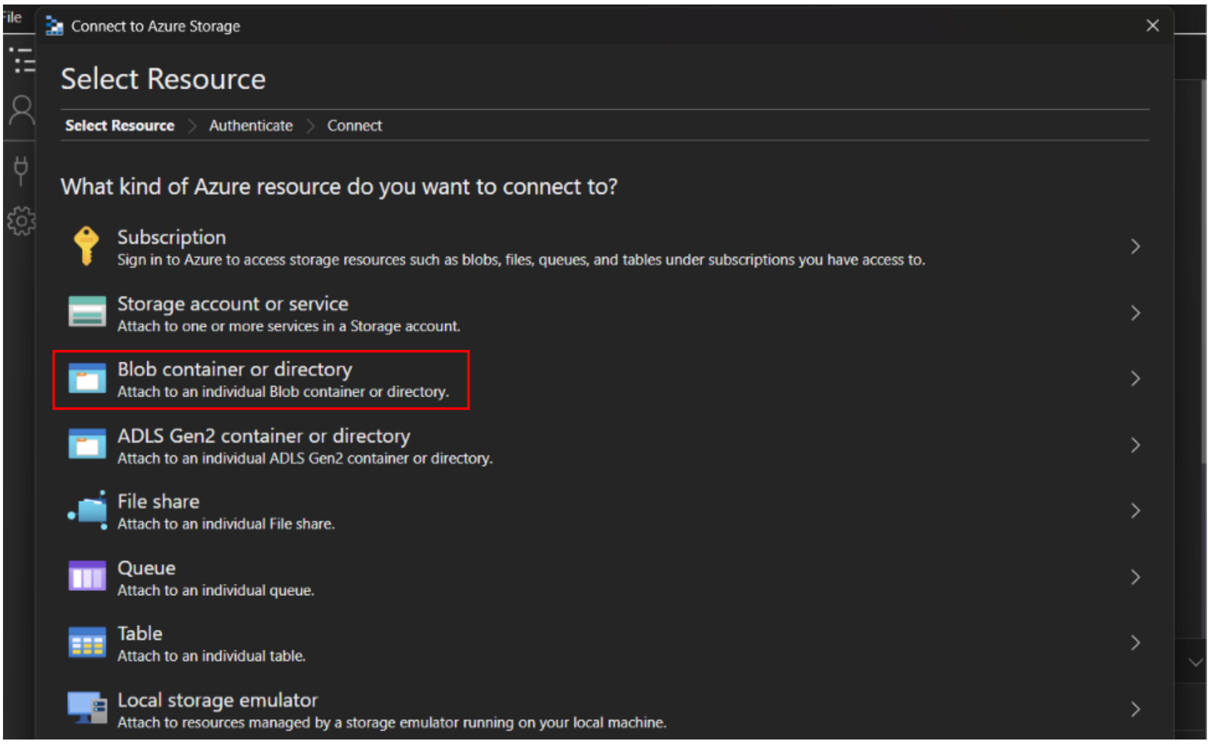
- Launch **Azure Storage Explorer**.
- Storage accounts(Right click).



- Select **Open Connect Dialog**.



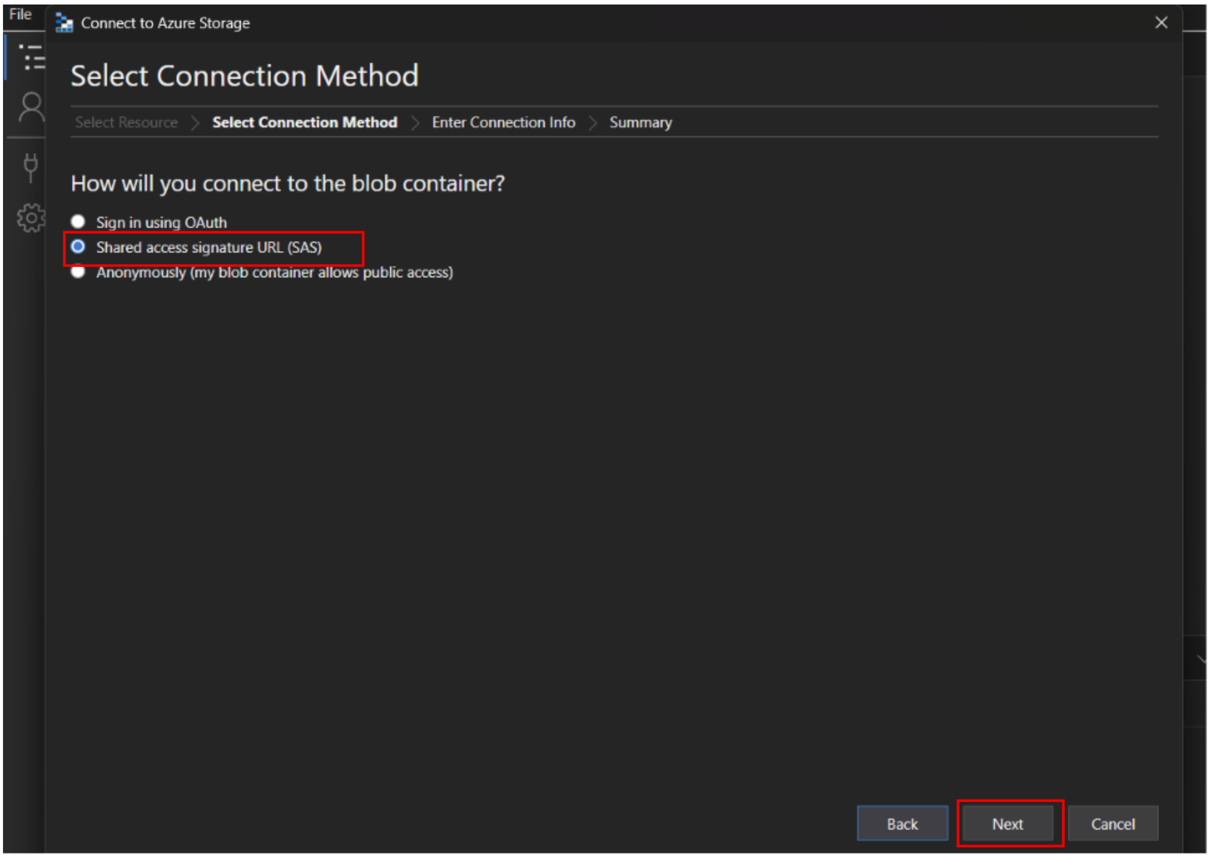
- Choose **Blob Container**.



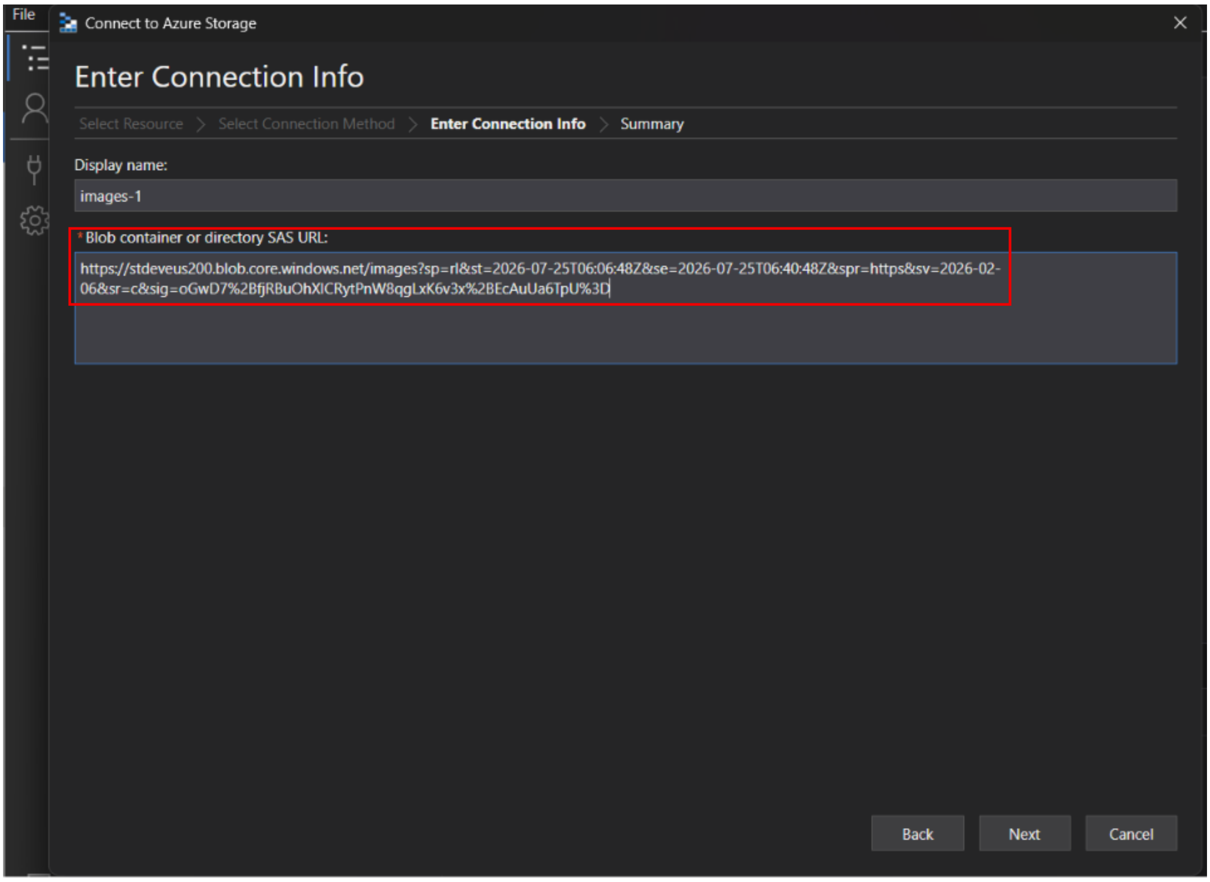
- Click **Next**.

Step 8: Connect Using the SAS URL

- Select **Shared Access Signature (SAS) URI** as the connection method.



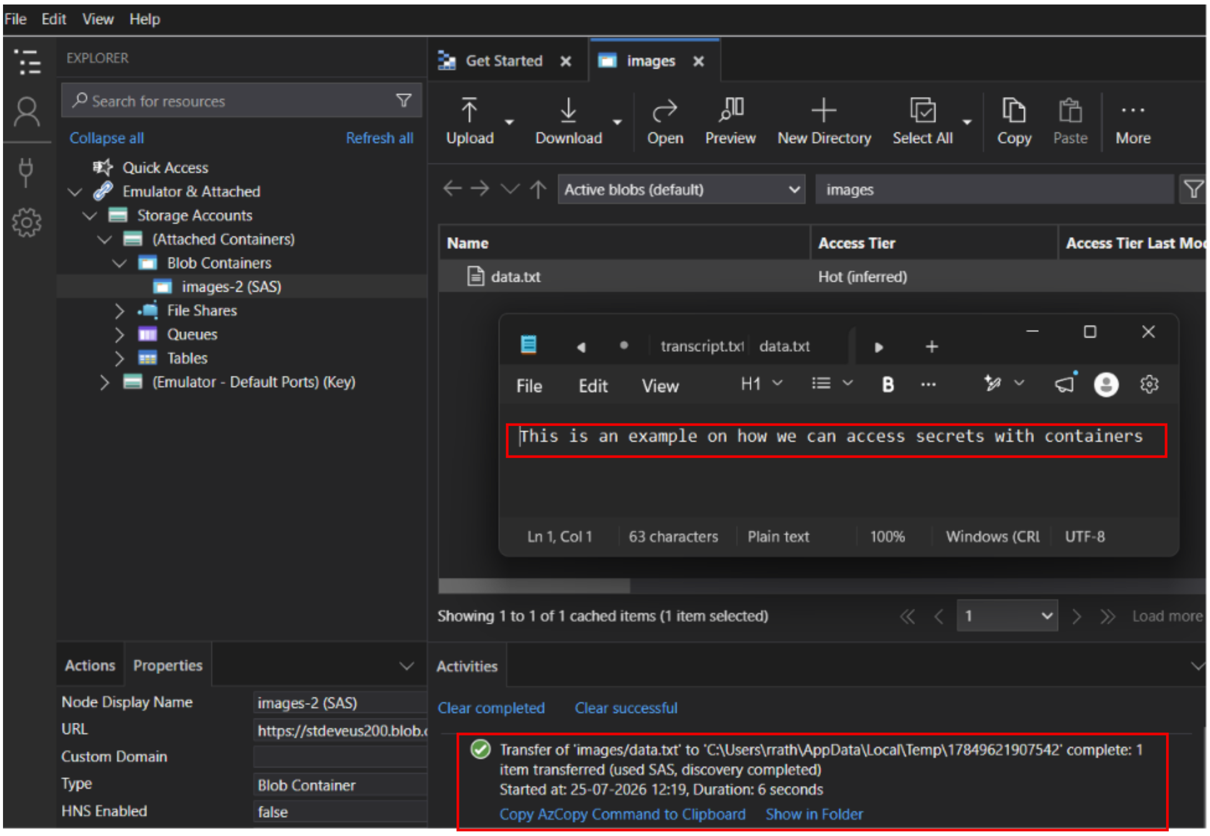
- Paste the copied **Container SAS URL**.



- Click **Next**.
- Review the connection details.
- Click **Connect**.

Step 9: Access the Container

- The container appears in Azure Storage Explorer.
- Browse the blobs inside the container.
- Perform only the operations allowed by the assigned permissions.
- Perform the **Read Permission**.



Step 10: Verify SAS Expiration

- Access the container before the expiry time.
- After the expiry time, try to reconnect or access the container.
- Azure denies access because the SAS has expired.